

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2025
Programme: BCA Full Marks : 100
Course: Microprocessor and Computer Architecture (New) Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define microprocessor system. Differentiate between microprocessor and microcontroller with appropriate block diagram. 7
- b) What is flag? Discuss about 8085 associated flags. Explain the addressing modes of 8085 microprocessor with examples. 8
2. a) Define Machine cycle. Draw well labelled timing diagram of instruction IN 20H. Calculate total execution time when operating frequency is 20KHz. 7
- b) Write an ALP in 8085 to transfer 8 bytes of data starting from memory location 3050H to 4050H. Comment in each line indicates full marks. 8

OR

Write an ALP in 8085 to calculate numbers of 1s available in data byte stored in memory location 2050H.

3. a) Draw and explain the internal architecture of 8086 microprocessor. 8
- b) Describe register organizations and data path available in computer system. 7
4. a) Differentiate between Control memory and Main memory. Explain the selection of address for control memory with its block diagram. 8
- b) Draw the block diagram of Microprogram Sequencer for a control memory, explain its operations. 7
5. a) Draw flow chart of Booth's multiplication algorithm. Perform $(-12) \times (4)$ using Booth's algorithm. 7
- b) Define Cache memory. Explain Cache Mapping Techniques. 8

OR

Design an address decoding circuit to interface 4KB ROM and 2KB RAM. The starting address is 5000H. Use suitable decoder.

6. a) How we can send data bytes from floppy disk to Main memory using DMA operation? Explain with proper block diagram. 7
- b) What do you mean by multicore architecture? Explain Flynn's taxonomy with suitable diagram. 8
7. Write short notes on: (Any two) 2×5
- a) 2's complement representation
- b) Interrupts
- c) Signed multiplication algorithm

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Attempt all the questions.

1. a) Differentiate between Microprocessor and Microcontroller with examples. 7
- b) What are the different types of FLAGS used in 8085 microprocessor? Show the status for the Flags after performing the following operation 5F H+ 02 H. 8
2. a) What is machine cycle? Explain the timing diagram of an instruction LXI H, 3050H. 7
- b) Write a 8085 ALP to find the largest number from an array of given 10 numbers stored at location starting from C000H and store the result in the memory location 2024H. 8

OR

Write a 8085 ALP to find smallest of three number.

3. a) Write an ALP to find the smallest number in a data array. Data from location C000H to C005H. 7

OR

Explain ALP development tools in 8086 microprocessor.

- b) What There is a very simple CPU for the given set of Instructions: 8

Instructions	Instruction Code	Operations
OR	00AAAAAA	AC ← AC ∨ M[AAAAAA]
AND	01AAAAAA	AC ← AC ∧ M[AAAAAA]
CMP	10AAAAAA	AC ← AC -
LDA	11XXXXXX	AC ← M[AAAAAA]

Let the instruction width be 8-bits and address is 6 bits. Design the CPU's Register Section, State Diagram and ALU.

4. a) Design the hardwired control unit for the CPU described in question 3 b. 7

- b) Design a micro-sequencer control unit with horizontal microcode for the given CPU in question number 3b. 8
5. a) Trace the multiplication of 4 and -7 using RTL code of Booth's algorithm. Also verify the obtained product. 8
- b) Explain the Set- Associative technique during the mapping of Cache memory and compare it with associative mapping. 7
6. a) How DMA reduces the CPU engagement dealing I/O interrupts with DMA than programmed I/O and interrupt driven I/O. Explain. 7
- b) Describe Flynn's Taxonomy. Explain the different topologies used in multiprocessor communication. 8
7. Write short notes on: (Any two) 2×5
 - a) RISC Vs. CISC
 - b) Hardwired Vs. micro programmed CU
 - c) Computer Organization Vs. Computer Architecture

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Attempt all the questions.

1. a) Difference between Von-Newmann and Harvard architecture with necessary diagram. 7
- b) What are the different types of FLAGS used in 8085 microprocessor? Show the status for the Flags after performing the following operation 5F H+ 02 H. 8
2. a) Explain the timing diagram for ORI 07H. 8
- b) Write an ALP for the following addition. 7
 $1^2+2^2+3^2+4^2+5^2+6^2+7^2+8^2+9^2$
3. a) What are the addressing modes of 8086 microprocessor? Explain in detail. 7

OR

Explain ALP development tools in 8086 with examples.

- b) Design a very simple CPU with given instruction set: 8

Instruction	Instruction Code	Micro-operations
ADD	00 AAAAAA	$AC \leftarrow AC + M[AAAAAA]$
JMP	01 AAAAAA	$PC \leftarrow AAAAAA$
OR	10 AAAAAA	$AC \leftarrow AC \vee M[AAAAAA]$
SKIP	11 XXXXXX	$PC \leftarrow PC + 1$

Write down the RTL codes for the CPU, draw state diagram, and register section and ALU.

4. a) Design a hardwired control unit for the CPU given in question number 3b. 7
- b) Design a micro-sequencer control unit with horizontal microcode for the given CPU in question number 3. b. 8

5. a) Draw the flowchart for the Booth's algorithm and perform -9×-9 . 7

OR

Show the division of the number 14/2 using Restoring Division algorithm.

- b) What is memory hierarchy? Differentiate between direct, associative and set associative mapping. 8
6. a) Draw and explain the flow of Interrupt I/O during input/output transfer. How is it more efficient than programmed I/O? 7
- b) What are the major conflicts occurring due to instruction pipelining in RISC? Explain. Describe the solutions to correct data conflicts. 8
7. Write short notes on: (Any two) 2×5
 - a) Flynn's taxonomy
 - b) HDMI
 - c) Instruction set

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Attempt all the questions.

1. a) How is microprocessor different from microcontroller in architecture? List out the applications. 7
- b) Draw the internal architecture of 8085 microprocessor and explain the registers from the architecture. 8
2. a) Write an assembly language program of 8085 to find the square of the data byte stored at memory location 3000H. 8

OR

Write a program to transfer 10 bytes of data stored at memory location 3000H to 3050H.

- b) Draw and explain the timing diagram for LDAX B, 2025H. 7
3. a) What do you understand by ALP development tools? Compare the addressing modes of 8086 microprocessor with 8085 microprocessor. 7
- b) A CPU having registers as very simple CPU has the following instruction set. Show the state diagram, register section and ALU. 8

Instruction	Mnemonic	Instruction code	Operation
Subtraction	SUB	00AAAAAA	$AC \leftarrow AC + M[AAAAAA]$
Logical AND	AND	01AAAAAA	$AC \leftarrow AC \wedge M[AAAAAA]$
Jump	JMP	10AAAAAA	GOTOAAAAAA
Decrement	DEC	11XXXXXX	$AC \leftarrow AC - 1$

OR

Explain the systematic process to design very simple CPU of your choice.

4. a) What do you mean by microinstruction sequencing? Explain. The process to generate microinstruction using horizontal and vertical microcode. 8
- b) Draw the flowchart for the Booth's algorithm and perform -8×-10 . 7
5. a) Draw the memory hierarchy pyramid and state the tradeoff. What is a cache memory? Explain different types of cache mapping techniques. 7
- b) Draw and explain the flow of DMA Transfer along with the signals used during the transfer. 8
6. a) What do you mean by polling in Programmed I/O? Explain how DMA controller facilitate in I/O operation with microprocessor. 7
- b) What is instruction pipeline? Differentiate between RISC and CISC architecture. 8
7. Write short notes on: (**Any two**) 2×5
 - a) Data conflicts and solution in Pipeline conflicts
 - b) Hardwired vs microprogrammed CU
 - c) Lookup ROM

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Attempt all the questions.

1. a) Define microcontroller. Differentiate between Harvard architecture and Von Newman architecture with their implementation. 7
- b) What are instruction sets? Explain instruction set of 8085 microprocessor with example. 8
2. a) Write an 8085 ALP to find the largest number from given array of 10 different numbers stored at memory location starting from 3200H and store it in the memory location 4200H. 8

OR

Write an 8085 ALP to divide a number 08H by 03H using immediate addressing mode to load register and store the result remainder to memory location 2300H and quotient to memory location 2301H.

- b) What do you understand by ALP development tools? Compare the addressing modes of 8086 microprocessor with 8085 microprocessor. 7
3. a) For a very simple CPU with the following instruction sets. Write down RTL, draw state diagram and design ALU and Register Set. 7

Instruction	Operation	Instruction code
ADD	$AC \leftarrow AC + M[AAAAAA]$	00AAAAAA
SUB	$AC \leftarrow AC - M[AAAAAA]$	01AAAAAA
OR	$AC \leftarrow AC \vee M[AAAAAA]$	10AAAAAA
SKIP	$PC \leftarrow PC + 1$	11XXXXXX

- b) Design a hardwired control unit with control signal generated for the CPU designed in question 3 (a). 8
4. a) Explain the microinstruction sequencing process to generate microinstruction using horizontal and vertical microcode. 7

- b) Compare Microprogrammed and Hardwired control unit design architecture and mention the merits and demerits of both. 8
5. a) Write down the RTL of Booth's Algorithm multiplication and verify it by performing 8×-7 . 8
- b) What do you mean by Look up ROM? Implement Look Up ROM for the combinational circuit given by function $f = x'yz + xy'z$. 7

OR

Implement Wallace tree to perform 3-bit multiplication using carry save adder.

6. a) What techniques are used for FIFO, LRU and LFU during cache replacement algorithms? Explain. 7
- b) What do you mean by polling in Programmed I/O? Explain how DMA controller facilitate in I/O operation with microprocessor. 8
7. Write short notes on: (**Any two**) 2×5
 - a) Memory Hierarchy
 - b) Instruction Pipeline Hazards
 - c) Flags of 8085